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1 =====
2   EVENT MANAGEMENT APP – DEPLOYMENT GUIDE (Vite + React)
3   Frontend Port: 5173 | Backend Port: 5000
4 =====
5
6 -----
7 STEP 1: LAUNCH EC2 INSTANCE
8 -----
9 - Go to AWS Console → EC2 → Launch Instance
10 - Choose Ubuntu (22.04 LTS recommended)
11 - Instance type: t2.micro (free tier) or higher
12 - Key pair: Create or select existing (.pem file)
13 - Keep default storage (8GB is fine)
14 - Launch the instance
15
16 -----
17 STEP 2: CREATE SECURITY GROUP (Inbound Rules)
18 -----
19 Add the following inbound rules:
20
21   Type          Protocol  Port    Source
22   -----
23   SSH           TCP       22      0.0.0.0/0
24   HTTP          TCP       80      0.0.0.0/0
25   Custom TCP    TCP      5000    0.0.0.0/0  ← Backend
26   Custom TCP    TCP      5173    0.0.0.0/0  ← Frontend (Vite)
27
28 NOTE: Use 0.0.0.0/0 for all CIDR blocks.
29 Attach this security group to your instance.
30
31 -----
32 STEP 3: CONNECT TO YOUR INSTANCE VIA SSH
33 -----
34 chmod 400 your-key.pem
35 ssh -i your-key.pem ubuntu@YOUR_VM_IP
36
37 -----
38 STEP 4: UPDATE SYSTEM & INSTALL NODE + NPM
39 -----
40 sudo apt update -y
41 sudo apt upgrade -y
42 sudo apt install nodejs npm -y
43
44 Verify installation:
45 node -v
46 npm -v
47
48 NOTE: If Node version is too old, install via NodeSource:
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49     curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
50     sudo apt install -y nodejs
51
52     -----
53 STEP 5: CLONE THE REPOSITORY
54     -----
55 git clone https://github.com/kaustubhgharat/Event_Management
56 cd ~/Event_Management
57
58     -----
59 STEP 6: CONFIGURE FRONTEND ENV FILE
60     -----
61 nano ~/Event_Management/frontend/.env
62
63     Change this line:
64     VITE_API_URL=http://localhost:5000
65
66     To:
67     VITE_API_URL=http://YOUR_VM_IP:5000
68
69 Save: Ctrl + O → Enter → Ctrl + X
70
71 IMPORTANT: Replace YOUR_VM_IP with your actual EC2 public IP address.
72 NOTE: Vite uses VITE_ prefix for env variables (not REACT_APP_).
73
74     -----
75 STEP 7: (NO EXTRA STEP NEEDED – handled in Step 9)
76     -----
77 Vite by default only serves on localhost. To expose it on the
78 network, just use the --host flag when starting (see Step 9).
79 No changes to vite.config.js required.
80
81     -----
82 STEP 8: START BACKEND
83     -----
84 cd ~/Event_Management/backend
85 npm install
86 node app.js
87
88 OPTIONAL – Use nodemon for auto-restart on file changes:
89     npm install -g nodemon
90     nodemon app.js
91
92 NOTE: Keep this terminal open. Open a new SSH session for frontend.
93
94     -----
95 STEP 9: START FRONTEND (New Terminal / SSH Session)
96     -----
```

```
97 cd ~/Event_Management/frontend
98 npm install
99 npm run dev -- --host
100
101 NOTE: Vite uses "npm run dev" – not "npm start".
102 NOTE: "--host" tells Vite to bind to 0.0.0.0 (all network interfaces)
103 so your app is reachable via public IP. Without it, only localhost works.
104
105 After running, Vite will show:
106   Local:   http://localhost:5173/
107   Network: http://YOUR_VM_IP:5173/   ← use this in browser
108
109 -----
110 STEP 10: TEST IN BROWSER
111 -----
112 Open: http://YOUR_VM_IP:5173
113
114 -----
115 TROUBLESHOOTING: RAM / Memory Issues
116 -----
117 If npm install or npm run dev crashes or hangs, add swap space:
118
119   sudo fallocate -l 1G /swapfile
120   sudo chmod 600 /swapfile
121   sudo mkswap /swapfile
122   sudo swapon /swapfile
123   free -h
124
125 OPTIONAL: To make swap permanent across reboots:
126   echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
127
128 TIP: Try running frontend first, then backend. If it still hangs,
129 apply the swap fix above and retry.
130
131 -----
132 TROUBLESHOOTING: Port Already in Use
133 -----
134 If port 5000 or 5173 is busy:
135   sudo lsof -i :5173
136   sudo kill -9 <PID>
137
138 -----
139 TROUBLESHOOTING: npm not found / old version
140 -----
141   node -v      ← should be 16+ ideally
142   npm -v
143   sudo npm install -g npm@latest
144
```

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145 -----
146 TROUBLESHOOTING: ERR_CONNECTION_REFUSED on browser
147 -----
148 Most likely cause: --host flag not used when starting Vite
149 → Stop and restart with: npm run dev -- --host
150
151 Also check: Is port 5173 open in your Security Group? (Step 2)
152
153 =====
154     QUICK REFERENCE
155 =====
156     Backend URL : http://YOUR_VM_IP:5000
157     Frontend URL: http://YOUR_VM_IP:5173
158     .env file   : ~/Event_Management/frontend/.env
159     Env variable: VITE_API_URL
160     Start cmd   : npm run dev -- --host (NOT npm start)
161 =====
162
```